Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

The purpose of this vulnerability assessment is to identify and evaluate the security risks associated with the company's publicly accessible database server. The database is highly valuable because it stores customer information and supports daily business operations for remote employees worldwide. Securing the server is important to protect sensitive data from unauthorized access, theft, or manipulation that could damage the company's reputation and finances. If the server were disabled or compromised, employees may lose access to customer data, disrupting operations, reducing productivity, and potentially causing loss of revenue and customer trust.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Standard user (Customer)	*Alter or delete critical information*	*1*	*3*	*3*
Standard user (Employee)	*Obfuscate future attacks*	*2*	*3*	*6*
Outsider (Hacker)	*Obtain sensitive information via exfiltration*	*3*	*3*	*9*

Approach

The three threats identified are hackers, malware or ransomware infections, and improper employee access. These risks are important because the publicly accessible database server is more vulnerable to attacks and unauthorized use. Such incidents could result in data breaches, operational interruptions, financial damage, and loss of customer trust.

Remediation Strategy

The company should implement stronger access controls to protect the database server from unauthorized users. Multi-factor authentication, role-based access control, and strong password policies can reduce the risk of external attacks and insider misuse. Replacing older SSL configurations with updated TLS encryption and using a VPN or IP allow-listing can secure remote connections. Regular system updates, malware protection, and activity logging should also be used to detect and prevent ransomware or other cyberattacks.